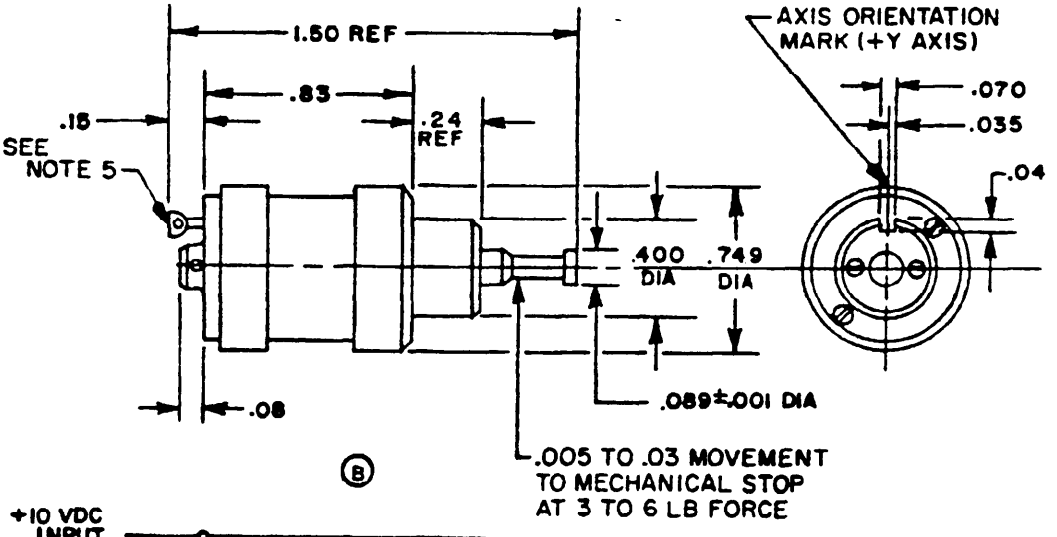
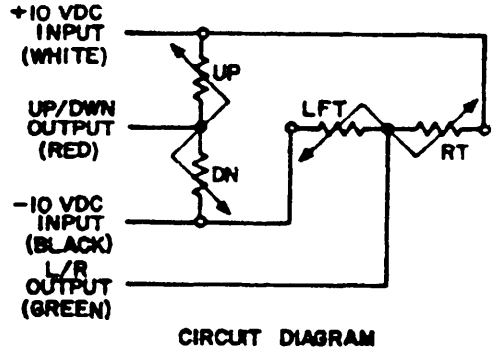
MILITARY SPECIFICATION SHEET

SWITCH, SOLID STATE TRANSDUCER,
2-AXIS JOYSTICK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-81619.




CIRCUIT DIAGRAM

| INCHES | MM |
|---|---|
| .001 | .03 |
| .0057 | .145 |
| .03 | .8 |
| .035 | .89 |
| .04 | 1.0 |
| .070 | 1.78 |
| .08 | 2.0 |
| .089 | 2.26 |
| .15 | 3.8 |
| .24 | 6.1 |
| .400 | 10.16 |
| .749 | 19.02 |
| .83 | 21.1 |
| 1.50 | 38.1 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Identification placed on body surface.
4. Unless otherwise specified, tolerances are ±.01 inch (.3 mm) on decimals and ±.5° on angles.
5. Flattened and pierced terminals equally spaced with 4 wire leads soldered in place. Wire leads shall be a minimum of 36" long and color coded per circuit diagram.

FIGURE 1. Configuration and dimensions.

Ⓑ denotes changes





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erein and in accordance with

3.2 Drawings. The solenoid requirements shall be as specified h

REQUIREMENTS:

OPERATIONS AND DIMENSIONS

Dimensions: See figure.
(B) Operating force: 3.1 pounds maximum (output proportional to force at a rate of .45 Vdc +20% per pound of force on either axis).
Pretravel: Not applicable.
Movement differential: Not applicable.
Overtravel: Not applicable.
(B) Terminals: Flexible wire (MIL-W-22759/7, P/N M22759/7-22).
Weight: 0.08 pound maximum.
Part Number: M81619/07-001.

MECHANICAL

Strength of actuating means: 15.0 pounds each direction along two perpendicular axes.
Strength of mounting means: Not applicable.
Terminal strength: 5.0 pounds (tensile).
Resistance to soldering heat: Not applicable.
Solderability: Not applicable.
Classification: Class 2 equipment per MIL-E-5400.
Enclosure type: Type I, unsealed.

ELECTRICAL

Resolution: Infinite.
Null temperature coefficient: ±.04% full scale per degree C.
Null hysteresis: ±.5% full scale within 1 second after release of shaft.
Sensitivity temperature coefficient: ±.2% full scale per degree C.
Resistance matching: 1% between each pair of elements.
(B) Absolute supply voltage: 12 volts dc maximum. (load impedance 20,000 ohms ±5%).
Absolute supply current: 30 mA maximum.
Supply voltage: 20 Vdc (±10 Vdc Bi-Polar).
Output voltage and current: .45 Vdc ±20% per pound force.
1.54 V maximum 20 mA maximum.
Load impedance: Not applicable.
Output: Rise time: Not applicable.
Fall time: Not applicable.
Output voltage waveform: Not applicable.
Short circuit: Applicable.
Transient voltage: Applicable.
Insulation resistance: Applicable.
Dielectric withstanding voltage: Applicable.



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ENVIRONMENTAL

Temperature
  Operating: -54°C to +71°C.
  Storage: -54°C to +71°C.
Shock
  Mechanical: 30 g for 11 milliseconds (no inspection during test).
  Temperature: -54°C and +71°C.
Vibration: Applicable (no inspection during test).
Acceleration: 10 g (no inspection during test).
Salt fog: Applicable.
Humidity: Applicable.
Sand and dust: Applicable.
Electromagnetic compatibility: Not applicable.
Magnetic compatibility: Not applicable.
Temperature-altitude: Applicable (no inspection during test).
  Altitude: Sea level to 70,000 feet.
Life: 150,000 cycles to maximum travel and return (no inspection during test).
Conditioning: Not applicable.

Custodian:
  Air Force - 11
  Army - EL
  Navy - AS

Review activity:
  Air Force - 85, 99
  Army - MI
  DLA - ES

Preparing activity:
  Air Force - 11

Agent:
  DLA - ES

(Project 5930-1074-3)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-S-81619/5B

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

C16061

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